

gam in HC	gam in FB	Trees	Trees in HC	Improved Index
$\{0, 0, 1, \frac{1}{2}\}$	$\{0, 0, 1, \frac{1}{2}\}$	$\{\text{Ch}[1, 0][1], \text{Ch}[1, 1]\}$	$\{\text{ChHC}[1, -1][1], \text{ChHC}[1, 0]\}$	1
$\{0, 1, -1, 0\}$	$\{0, 1, 0, 0\}$	$\{\text{Ch}[0, 0][1], \text{Ch}[1, 0]\}$	$\{\text{ChHC}[0, 0][1], \text{ChHC}[1, -1]\}$	$-y - \frac{1}{y} + O\left[\frac{1}{y}\right]^{101}$
$\{0, 1, 0, \frac{1}{2}\}$	$\{0, 1, 1, \frac{1}{2}\}$	$\{\text{Ch}[0, 0][1], \text{Ch}[1, 1]\}$	$\{\text{ChHC}[0, 0][1], \text{ChHC}[1, 0]\}$	$y^2 + 1 + \left(\frac{1}{y}\right)^2 + O\left[\frac{1}{y}\right]^{101}$
$\{0, 2, -1, \frac{1}{2}\}$	$\{0, 2, 1, \frac{1}{2}\}$	$\{\text{Ch}[1, 1], \{2 \text{Ch}[0, 0][1], \text{Ch}[1, 0]\}\}$	$\{\text{ChHC}[1, 0], \{2 \text{ChHC}[0, 0][1], \text{ChHC}[1, -1]\}\}$	$y^4 + y^2 + 1 + \left(\frac{1}{y}\right)^2 + \left(\frac{1}{y}\right)^4 + O\left[\frac{1}{y}\right]^{101}$
$\{0, 2, 0, 0\}$	$\{0, 2, 2, 0\}$	$\{\text{Ch}[1, 1], \text{Ch}[-1, -1][1]\}$	$\{\text{ChHC}[1, 0], \text{ChHC}[-1, 0][1]\}$	$-y^5 - y^3 - y - \frac{1}{y} - \left(\frac{1}{y}\right)^3 - \left(\frac{1}{y}\right)^5 + O\left[\frac{1}{y}\right]^{101}$
$\{0, 3, -1, 0\}$	$\{0, 3, 2, 0\}$	$\{\text{Ch}[-1, -1][1], \{\text{Ch}[1, 1], \{\text{Ch}[0, 0][1], \text{Ch}[1, 0]\}\}\}$	$\{\text{ChHC}[-1, 0][1], \{\text{ChHC}[1, 0], \{\text{ChHC}[0, 0][1], \text{ChHC}[1, -1]\}\}\}$	$-y^9 - 3y^7 - 4y^5 - 4y^3 - 4y - \frac{4}{y} - \frac{4}{y^3} - \frac{4}{y^5} - \frac{3}{y^7} - \left(\frac{1}{y}\right)^9 + O\left[\frac{1}{y}\right]^{101}$
$\{0, 3, 0, \frac{1}{2}\}$	$\{0, 3, 3, \frac{1}{2}\}$	$\{\text{Ch}[-1, -1][1], \{\text{Ch}[0, 0][1], 2 \text{Ch}[1, 1]\}\}$	$\{\text{ChHC}[-1, 0][1], \{\text{ChHC}[0, 0][1], 2 \text{ChHC}[1, 0]\}\}$	$y^{10} + 2y^8 + 3y^6 + 3y^4 + 3y^2 + 3 + \frac{3}{y^2} + \frac{3}{y^4} + \frac{3}{y^6} + \frac{2}{y^8} + \left(\frac{1}{y}\right)^{10} + O\left[\frac{1}{y}\right]^{101}$
$\{0, 4, -2, 0\}$	$\{0, 4, 2, 0\}$	$\{\text{Ch}[-2, -1][1], \text{Ch}[2, 1], \{\text{Ch}[-1, -1][1], \{\text{Ch}[1, 1], \{2 \text{Ch}[0, 0][1], 2 \text{Ch}[1, 0]\}\}\}\}$	$\{\text{ChHC}[-2, 1][1], \text{ChHC}[2, -1], \{\text{ChHC}[-1, 0][1], \{\text{ChHC}[1, 0], \{2 \text{ChHC}[0, 0][1], 2 \text{ChHC}[1, -1]\}\}\}\}$	$-y^{13} - 3y^{11} - 8y^9 - 10y^7 - 11y^5 - 11y^3 - 11y - \frac{11}{y} - \frac{11}{y^3} - \frac{11}{y^5} - \frac{10}{y^7} - \frac{8}{y^9} - \frac{3}{y^{11}} - \left(\frac{1}{y}\right)^{13} + O\left[\frac{1}{y}\right]^{101}$
$\{1, 2, 0, 2\}$	$\{1, 2, 2, 2\}$	$\{\text{Ch}[2, 1], \{\text{Ch}[1, 0][1], \text{Ch}[1, 1]\}\}$	$\{\text{ChHC}[2, -1], \{\text{ChHC}[1, -1][1], \text{ChHC}[1, 0]\}\}$	1
$\{1, 0, 1, -\frac{1}{2}\}$	$\{1, 0, 1, -\frac{1}{2}\}$	$\{\text{Ch}[0, 0], \{\text{Ch}[-2, -2][1], \text{Ch}[-2, -1]\}\}$	$\{\text{ChHC}[0, 0], \{\text{ChHC}[-2, 0][1], \text{ChHC}[-2, 1]\}\}$	1
$\{1, 1, 1, 0\}$	$\{1, 1, 2, 0\}$	$\{\text{Ch}[1, 1], \{\text{Ch}[-2, -2][1], \text{Ch}[-2, -1]\}\}$	$\{\text{ChHC}[1, 0], \{\text{ChHC}[-2, 0][1], \text{ChHC}[-2, 1]\}\}$	1
$\{1, 2, 1, \frac{3}{2}\}$	$\{1, 2, 3, \frac{3}{2}\}$	$\{\{\text{Ch}[-2, -2][1], \text{Ch}[-2, -1]\}, \{\text{Ch}[2, 1], \{\text{Ch}[1, 0][1], \text{Ch}[1, 1]\}\}\}$	$\{\{\text{ChHC}[-2, 0][1], \text{ChHC}[-2, 1]\}, \{\text{ChHC}[2, -1], \{\text{ChHC}[1, -1][1], \text{ChHC}[1, 0]\}\}\}$	1
$\{-1, 0, 1, \frac{1}{2}\}$	$\{-1, 0, 1, \frac{1}{2}\}$	$\{\text{Ch}[0, 0][1], \{\text{Ch}[1, 0][1], \text{Ch}[1, 1]\}\}$	$\{\text{ChHC}[0, 0][1], \{\text{ChHC}[1, -1][1], \text{ChHC}[1, 0]\}\}$	1
$\{-1, -2, 1, -\frac{3}{2}\}$	$\{-1, -2, -1, -\frac{3}{2}\}$	$\text{Ch}[2, 1][1]$	$\text{ChHC}[2, -1][1]$	1
$\{1, 0, 0, -1\}$	$\{1, 0, 0, -1\}$	$\{\text{Ch}[-2, -2][1], 2 \text{Ch}[-1, -1]\}, \{\{\text{Ch}[-2, -2][1], \text{Ch}[-2, -1]\}, \{\text{Ch}[-2, -1][1], 2 \text{Ch}[-1, -1]\}\}$	$\{\text{ChHC}[-2, 0][1], 2 \text{ChHC}[-1, 0]\}, \{\{\text{ChHC}[-2, 0][1], \text{ChHC}[-2, 1]\}, \{\text{ChHC}[-2, 1][1], 2 \text{ChHC}[-1, 0]\}\}$	$y^2 + 2 + \left(\frac{1}{y}\right)^2 + O\left[\frac{1}{y}\right]^{101}$
$\{1, 0, 0, -2\}$	$\{1, 0, 0, -2\}$	$\{\text{Ch}[-1, -1], \{\text{Ch}[-3, -2][1], \text{Ch}[-2, -1]\}\}, \{\{\text{Ch}[-5, -4][1], \text{Ch}[-5, -3]\}, \{\text{Ch}[-2, -1][1], 2 \text{Ch}[-1, -1]\}\}, \{\{\text{Ch}[-3, -2][1], \text{Ch}[-2, -2]\}, \{\text{Ch}[-1, -1], \{\text{Ch}[-2, -2][1], \text{Ch}[-2, -1]\}\}\}$	$\{\text{ChHC}[-1, 0], \{\text{ChHC}[-3, 1][1], \text{ChHC}[-2, 1]\}\}, \{\{\text{ChHC}[-5, 1][1], \text{ChHC}[-5, 2]\}, \{\text{ChHC}[-2, 1][1], 2 \text{ChHC}[-1, 0]\}\}, \{\{\text{ChHC}[-3, 1][1], \text{ChHC}[-2, 0]\}, \{\text{ChHC}[-1, 0], \{\text{ChHC}[-2, 0][1], \text{ChHC}[-2, 1]\}\}\}$	$y^4 + 3y^2 + 6 + \frac{3}{y^2} + \left(\frac{1}{y}\right)^4 + O\left[\frac{1}{y}\right]^{101}$
$\{2, 1, 0, -\frac{1}{2}\}$	$\{2, 1, 1, -\frac{1}{2}\}$	$\{3 \text{Ch}[0, 0], \text{Ch}[-1, -1][1]\}$	$\{3 \text{ChHC}[0, 0], \text{ChHC}[-1, 0][1]\}$	1
$\{2, 1, -1, -1\}$	$\{2, 1, 0, -1\}$	$\{2 \text{Ch}[0, 0], \{\text{Ch}[-2, -1][1], \text{Ch}[-1, -1]\}\}$	$\{2 \text{ChHC}[0, 0], \{\text{ChHC}[-2, 1][1], \text{ChHC}[-1, 0]\}\}$	$-y - \frac{1}{y} + O\left[\frac{1}{y}\right]^{101}$